VICTORIA JUNIOR COLLEGE
BIOLOGY DEPARTMENT
JC2 PRELIMINARY EXAMINATIONS 2011
Higher 1

CANDIDATE NAME

CLASS | | | | | | INDEX NUMBER | | |

BIOLOGY 8875/02

Paper 2 Core Paper **12 September 2011**

Additional Materials: Answer Paper **2 hours**

READ THESE INSTRUCTIONS FIRST

Write your CLASS/ INDEX no. and name on all the work you hand in.
Write in dark blue or blue pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use any staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions.

Section B
Answer **one** question.
Write your answers on separate answer paper provided.

At the end of the examinations,
1. Hand in section A and the question you attempted from section B separately;
2. Fasten all your work securely;
3. Enter the number of the section B question you have answered in the grid opposite.

The intended number of marks is given in brackets [ ] at the end of each question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
Section B	
Total	

Section A

Answer **all** the questions in this section.

1 A research group discovers a hydrolytic enzyme in the aleurone of the barley seed that converts lipids to simpler molecules and begins to characterize it. The researchers found that the three essential catalytic groups in the active site are contributed by histidine 57, aspartate 102 and serine 195, where the numbers denote the position of the amino acid in the amino acid sequence.

(a) Describe the advantages of storing lipids rather than starch in a seed.

[2]

(b) Suggest with reasons where the hydrolytic enzyme is located in the aleurone cell.

[2]

(c) Explain what determines the precise position of these three amino acids in the structure of the hydrolytic enzyme.

[3]

The researchers discovered that the seed germinates best if the soil is cool at close to 5 ^{0}C. In an investigation into the properties of the hydrolytic enzyme, they set up three water baths at 15, 20 and 25 ^{0}C. Into each bath was placed a tube containing 1 cm^3 of the enzyme solution and a tube containing 10 cm^3 of concentrate substrate solution, on reaching the required temperature, the enzyme and substrate were quickly mixed and kept in the water bath.

There was a large excess of the substrate, so that the substrate concentration was not a limiting factor.

Samples were taken from each tube at regular intervals and the concentration of the product in these samples was determined. The results are shown in **Fig. 1.1**.

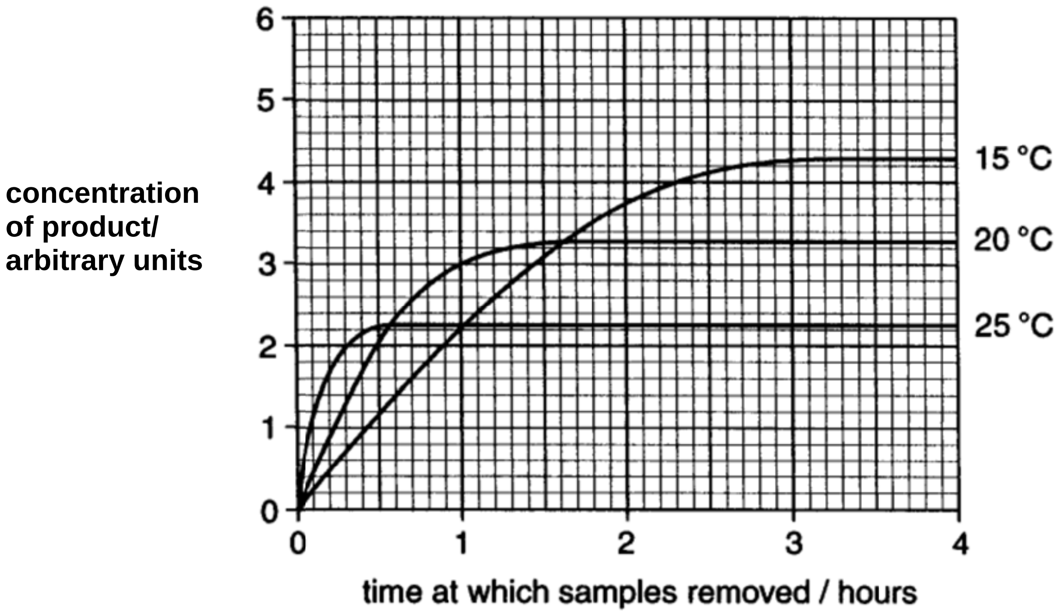


Fig. 1.1

(d) Account for the curve at 15 ^{0}C in **Fig. 1.1**.

[3]

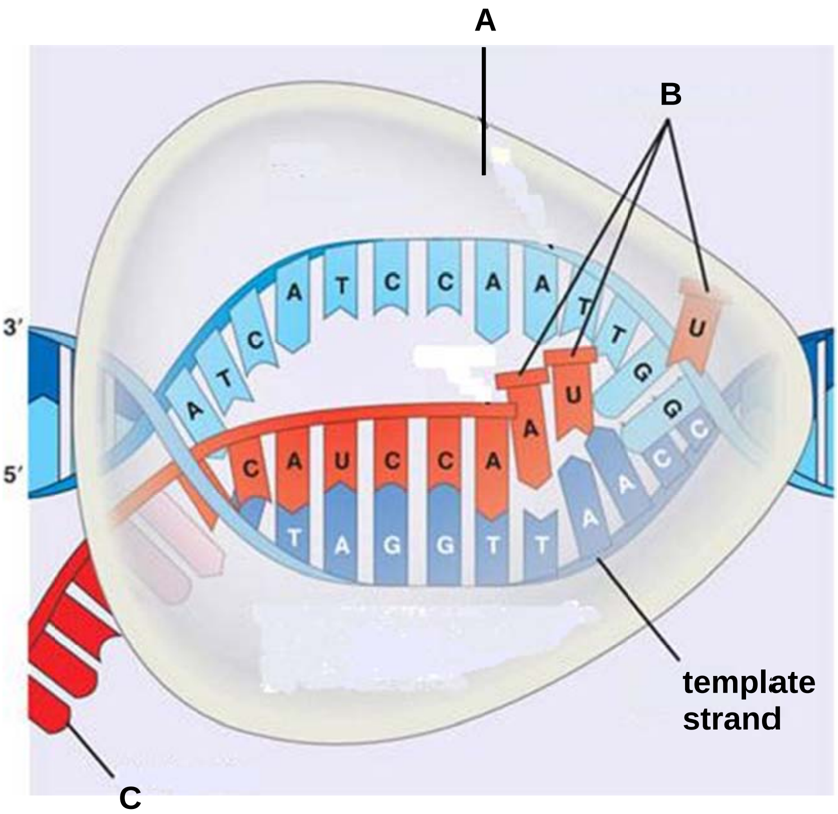
(e) Predict and explain the effect of carrying out the procedure at 5 ^{0}C. Sketch your prediction on **Fig. 1.1**. Label it as Y. [1]

Explanation:

[2]

[Total: 13]

2 **Fig. 2.1** is a diagram showing transcription in a eukaryotic cell, in which a section of the gene sequence is shown.



(Adapted from http://www.newsperuvian.com/dna/dna-transcription/)

Fig. 2.1

(a) Identify the structures **A** and **B**.

A ……………………………………………………………………………………………

B …………………………………………………………………………………………… [2]

(b) A more complete sequence of **C** is as shown in **Fig. 2.2**.

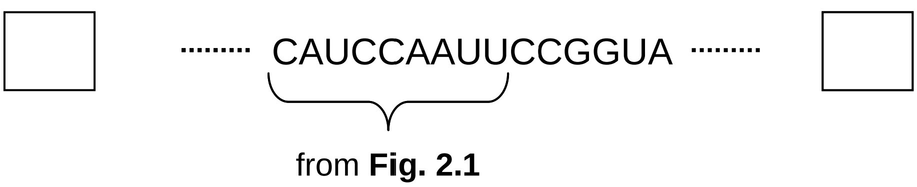


Fig. 2.2

(i) Indicate in the boxes above the 5’ and 3’ end of structure **C**. [1]

Second letter

First letter	U	C	A	G	Third letter
U	UUU } Phe UUC } UUA } Leu UUG }	UCU } UCC } Ser UCA } UCG }	UAU } Tyr UAC } UAA Stop UAG Stop	UGU } Cys UGC } UGA Stop UGG Trp	U C A G
C	CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } His CAC } CAA } Gln CAG }	CGU } CGC } Arg CGA } CGG }	U C A G
A	AUU } AUC } Ile AUA } AUG Met	ACU } ACC } Thr ACA } ACG }	AAU } Asn AAC } AAA } Lys AAG }	AGU } Ser AGC } AGA } Arg AGG }	U C A G
G	GUU } GUC } Val GUA } GUG }	GCU } GCC } Ala GCA } GCG }	GAU } Asp GAC } GAA } Glu GAG }	GGU } GGC } Gly GGA } GGG }	U C A G

(Taken from http://www.cbs.dtu.dk/staff/dave/Fig13_18.JPG)

Table 2.1

(ii) Using **Table 2.1**, write down the amino acid sequence translated from **Fig. 2.2**.

.... [1]

(iii) Explain how you have applied the features of the Genetic Code to derive your answer in **(ii)**.

....

....

....

....

....

.... [3]

(c) In eukaryotic cells, the mRNA transcribed undergoes RNA splicing to form mature mRNA.

Explain what happens if a mutation occurs at one of the splice site of an intron.

[3]

[Total: 10]

3 A type of seaweed *Uvla taeniata* was grown under light of varying wavelengths. **Fig. 3.1** shows the correlation of the absorption spectrum with the action spectrum (rate of photosynthesis) in the plant.

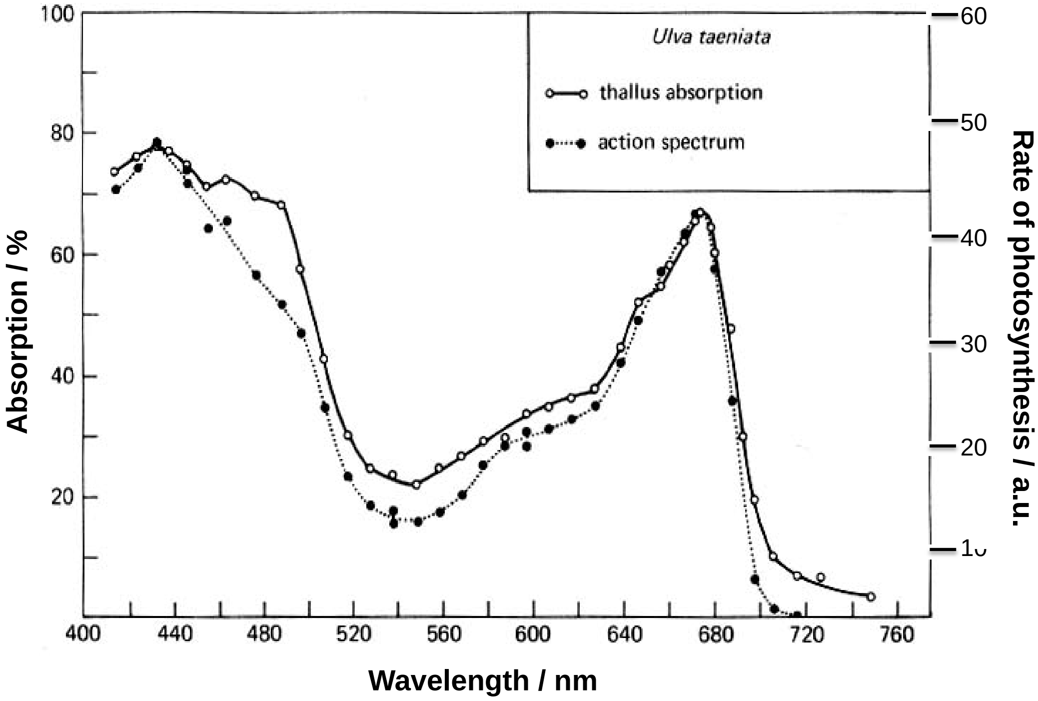


Fig. 3.1

(a) (i) Outline the role of light in light dependent reactions.

[2]

(ii) Suggest a parameter that can be used to measure the rate of photosynthesis in this experiment.

[1]

(iii) With reference to **Fig. 3.1**, deduce the wavelengths of light *Uvla taeniata* is most effective in bringing about photosynthesis. Explain your answer.

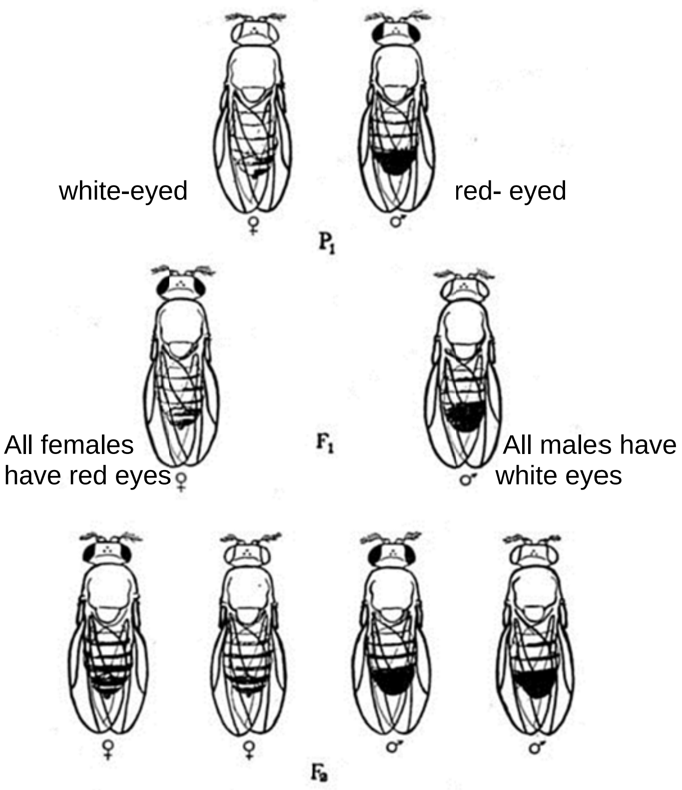
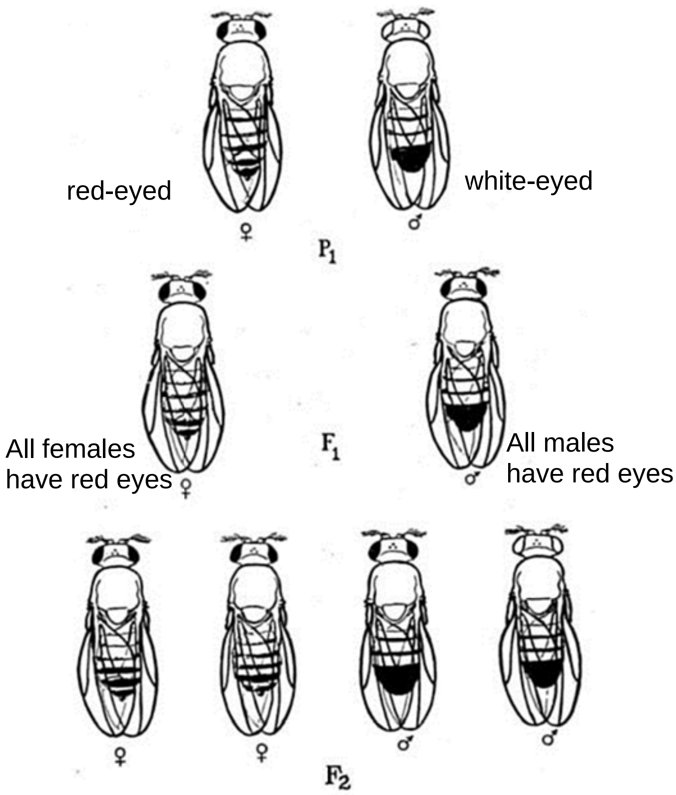
[2]

(b) A chemical compound, added to a chloroplast extract of *Uvla taeniata*, preferentially binds to free electrons released from P700 only. With reference to the light dependent reactions only, explain the effect of this inhibitor on the production of ATP and reduced NADP.

[2]

[Total: 7]

4 In an investigation into the inheritance of eye colour (red and white) in *Drosophila*, two separate crosses were carried out. In *Drosophila*, the female is the homogametic sex (XX) and the male the heterogametic sex (XY). In both crosses, pure bred flies were used. The results of the crosses are shown below.

<table>
<tr><th>Cross 1</th><th>Cross 2</th></tr>
<tr><td>
50% of the female flies have red eyes, 50% white eyes
50% of the male have red eyes, 50% white eyes.</td><td>
All the female flies have red eyes wheres 50% of the male have red eyes, 50% white eyes.</td></tr>
</table>

(a) Explain the significance of carrying out crosses 1 and 2.

....

....

.... [2]

(b) From the cross 2 above, state which eye colour is dominant and explain your answer.

....

.... [1]

(c) Using suitable symbols, construct a genetic diagram of Cross 1.

[6]

Studies of the white eye mutant flies reveal that the mutation does not affect the activity of the enzymes involved in pigment synthesis but instead interfere with the ability of the cells to take up the pigment precursors.

(d) Based on the information given, suggest what the normal allele codes for and how a mutation can result in the expression of the white eye phenotype.

[3]

[Total: 12]

5 Restriction digests were performed separately on isolated samples of a plasmid using three different restriction enzymes – R1, R2 and R3.

The products of each digestion reaction were added into the wells of a gel and an electric current passed through it. **Fig 5.1** shows the results obtained after 20 minutes. A DNA ladder was loaded into the well labelled D.

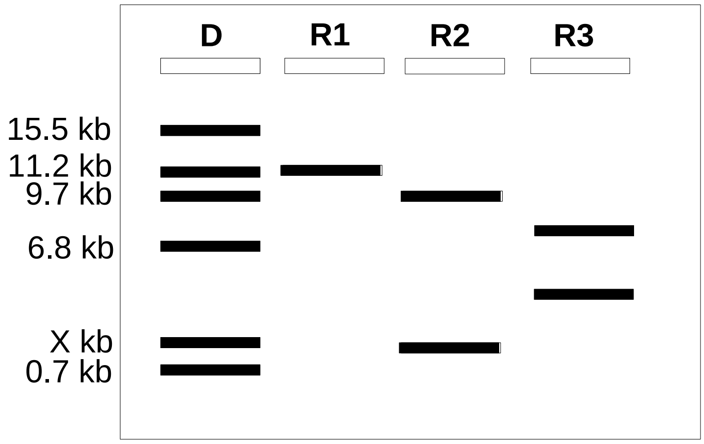


Fig. 5.1

(a) Explain the significance of the DNA ladder in a gel electrophoresis experiment.

....

....

.... [2]

(b) DNA fragments are invisible to the naked eye. State a way in which all the restriction fragments can be observed in this experiment.

....

.... [1]

(c) With reference to **Fig. 5.1** above,

(i) which restriction enzyme cuts the plasmid at only one site? Explain why.

....

....

.... [2]

(ii) deduce the unknown fragment size labeled X.

.... [1]

(iii) explain why R3 may not be suitable for use in genetic engineering experiment.

---

[2]

[Total: 8]

Section B

Answer EITHER 6 OR 7

Write your answers on the separate answer paper provided.
Your answers should be illustrated by clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in section **(a), (b)** etc., as indicated in the question.

Either

6 **(a)** Describe starting from mature mRNA, how a glycoprotein is synthesized and embedded in the cell surface membrane. [10]

(b) Explain the need for reduction division (meiosis) prior to fertilisation in sexual reproduction. [4]

(c) Explain how meiosis and random fertilisation can lead to variation. [6]

[Total: 20]

Or

7 **(a)** Using a named example each for a fibrous protein and a globular protein, explain how the differences in their structure allow them to perform their respective function. [8]

(b) Explain, with examples, how environmental factors act as forces of natural selection. [6]

(c) Comment on the need for production of genetically identical cells and fine control of replication. [6]

[Total: 20]